

TAIKI METOKI

AI for Drug Discovery | Medical Imaging | LLM Post-training

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[Portfolio](#) | [GitHub](#) | [Kaggle](#) | [Hugging Face](#) | [LinkedIn](#)

Daily base: AIST Tokyo Waterfront, 2-3-26 Aomi, Koto-ku, Tokyo 135-0064, Japan

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SUMMARY

Master's student working at the intersection of AI for drug discovery, medical imaging, and LLM post-training. Hands-on research and engineering experience across academia, national research institutes, and AI startups. Kaggle Competition Expert with two silver medals (2026 Vesuvius Challenge, 2025–26 CSIRO Image2Biomass), one poster presentation in computational chemistry, and contributions to large-scale Japanese LLM development through LLM-jp.

EDUCATION

Yokohama City University, Graduate School of Medical Life Science Apr 2026 – Present
M.Sc. candidate, Structural Cell Biology Lab (Advisor: Prof. Kentaro Tomii) Yokohama, Japan

- Research focus: AI for drug discovery, bioinformatics, and chemoinformatics with large language models.

Shinshu University, Faculty of Textile Science and Technology Apr 2022 – Mar 2026
B.Eng., Department of Chemistry and Materials (Advisor: Prof. Iori Shimada) Nagano, Japan

- Thesis: Deep-learning approaches to chemical reaction prediction.
- GPA: 3.39 / 4.00.

RESEARCH & INDUSTRY EXPERIENCE

LLM-jp (NII / large-scale Japanese LLM consortium) Oct 2025 – Present
Research Assistant (May 2026 – Present); Student Volunteer (Oct 2025 – Apr 2026) Tokyo, Japan

- Contributing to LLM post-training research, with a focus on reinforcement learning, reasoning, and domain adaptation.
- Collaborating with researchers across universities and NII on training and evaluation of open Japanese LLMs.

Elith, Inc. Oct 2025 – Present
Machine Learning Engineer Intern Tokyo, Japan

- Developing GENFLUX, a generative-flow-based machine learning system.

National Institute of Advanced Industrial Science and Technology (AIST) Jul 2025 – Present
Technical Trainee, AI for Drug Discovery Tokyo Waterfront, Japan

- Applying machine learning and bioinformatics methods to drug discovery problems under the AIST research program.

Cubec, Inc. May 2025 – Present
Data Scientist Intern Tokyo, Japan

- Developing medical-domain large language models for clinical applications.

Nara Institute of Science and Technology (NAIST) Feb 2025 – Mar 2025
Research Intern, Medical Image Lab Nara, Japan

- Estimated lumbar-spine bone mineral density from X-ray images using deep learning.

Hips, Inc. May 2024 – Aug 2024

- Performed customer-data analysis to support business decision-making.

SELECTED COMPETITIONS

Kaggle: Vesuvius Challenge — Surface Detection

Feb 2026

Silver Medal — 19 / 1391 (Top 1.4%)

- Developed nnU-Net-based detection models and built the post-processing pipeline for the final solution.

Kaggle: CSIRO — Image2Biomass Prediction

Nov 2025 – Jan 2026

Silver Medal — 133 / 3802 (Top 3.5%)

- Built DINOv3-based models and ran ~50 experiments for model selection and subtask validation.
[\[Solution write-up\]](#)

Kaggle: RSNA Intracranial Aneurysm Detection

Sep 2025 – Oct 2025

- Built a 2.5D baseline for aneurysm detection on medical imaging data; published the [solution repository](#).

Matsuo Lab LLM Development Competition 2025

Jul 2025 – Sep 2025

- Phase 1: Led SFT and GRPO training of Qwen3-based models.
- Phase 2: Engineered the training/evaluation environment for the runner-up team.

Kaggle: CZII — CryoET Object Identification

Dec 2024 – Feb 2025

101 / 931 (Top 11%)

- 3D object identification in cryo-electron tomography volumes for structural biology applications.

LLMATCH Season 1

Dec 2024 – Mar 2025

- Explored LLM-driven research workflows for drug discovery.

PUBLICATIONS & PRESENTATIONS

- T. Metoki, N. Yokosuka, I. Shimada. “*Extension of Chemical Reaction Neural Networks to Heterogeneous Catalytic Reaction Systems and Improvement of Prediction Accuracy.*” Poster Presentation, Society of Computer Chemistry, Japan (JCCJ), Nov 2025.

TECHNICAL SKILLS

- **Languages:** Python, Julia.
- **ML / DL:** PyTorch, Hugging Face Transformers, nnU-Net, DINOv3.
- **Domains:** Medical Imaging, LLM Post-training (SFT / RLHF / GRPO), Bioinformatics, Chemoinformatics, Scientific ML, Computer Vision.
- **Infrastructure:** Docker, HPC / GPU clusters, Git.
- **Languages (spoken):** Japanese (native), English (TOEIC 720, 2024).

SCHOLARSHIPS

- TAKEUCHI Scholarship Foundation (2024–2025).
- Hokushin Scholarship Foundation (2024–2025).
- Imon Scholarship Foundation (2024–2025).
- Sato Sadao International Scholarship Foundation (2024–2025).